Introduction to Machine Learning

Lesson 1 - What is Machine Learning?

What is machine learning?

• Automating automation

• Getting computers to programme themselves

• 60-year-old field of research

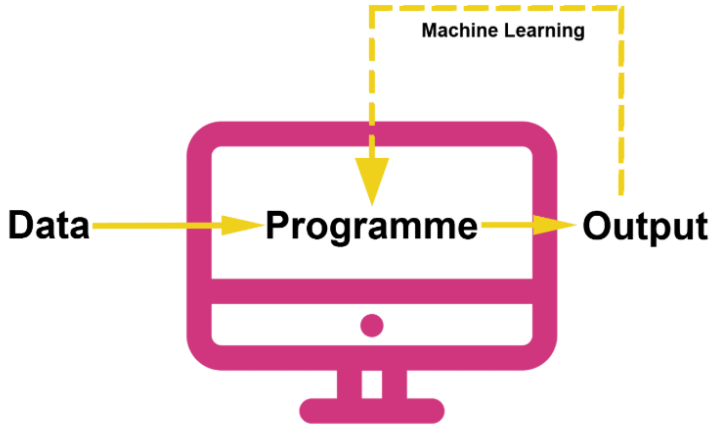
• "Let the data do the work"

Nowadays we hear a lot about automation, machine, algorithms, and artificial intelligence. Widely known as the **fourth industrial revolution**, machines have started to make decisions without the help of humans in different sectors. From manufacturing to driving cars or selecting the right life partner for individuals. The engine that is driving this revolution is machine learning where, like humans, machines learn from their environment and their mistakes. Interestingly, these machine learning algorithms were introduced many decades ago. However, the lack of big data and, in particular real-time big data, that we now can get from internet of things devices (IoT) and social media, stalled the application of these algorithms. Currently, machine learning is affecting every part of our lives and has started to change our behaviour and mindset. Most interestingly it has started to compete with the fundamental trait of humanity – creativity.

Lesson 2 - Software Engineering vs. Machine Learning

Excel and Tableau are programs that output various results from data analysis.

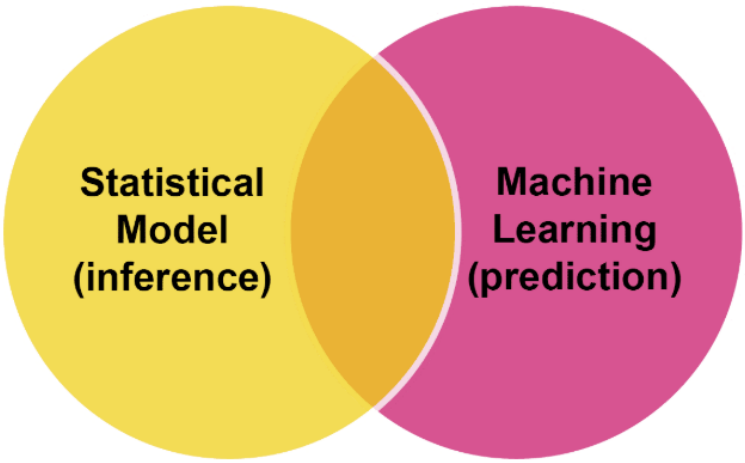
This includes visualisations (graphs, charts etc) for a given dataset.



A machine learning algorithm learns from its output and reprogrammes the program to give better results/prediction/classification with the next input data. This self-programming is what makes ML unique from software engineering.

Lesson 3 - Statistical Model vs. Machine Learning

Statistical modelling focuses on inference while machine learning focuses on prediction.



Statistical Model	Machine Learning
No training	Training
No adjustment	Adjustment
No testing	Testing

Lesson 4 - Big Data and Future Society

Big Data and Future Society.

- How Big is it?
- Why do all these companies care?
- Why should you care?

As big data has such a significant influence on machine learning, we need to learn more about it and its impact.

- How big is it?

Both the volume and the variety of data has grown exponentially over the last two decades.

- Why do all these companies care?

Every minute we are creating huge amounts of data and the tech giants like Google and Facebook are using this data for their commercial benefits. It is estimated that the US market revenue alone will be over 100 billion in next 5 years.

As we are moving towards a virtual society, our digital footprint in this virtual society is increasing at an exponential rate. However, the regulation around using this digital footprint are physical world centric. This gap creates the opportunity for exploitation.

- Your personality – is it what you are?

Our digital footprint not only defines us as a person but also our personality. From our digital footprint data, machine learning can easily define an individual's personality by learning from his or her big five personality traits, as described by the most prominent modern theory of personality traits - the Five Factor Model - conscientiousness, agreeableness, neuroticism, openness to experience, and extraversion (Vo et al., 2018). This is the basis for **Social Computing.**

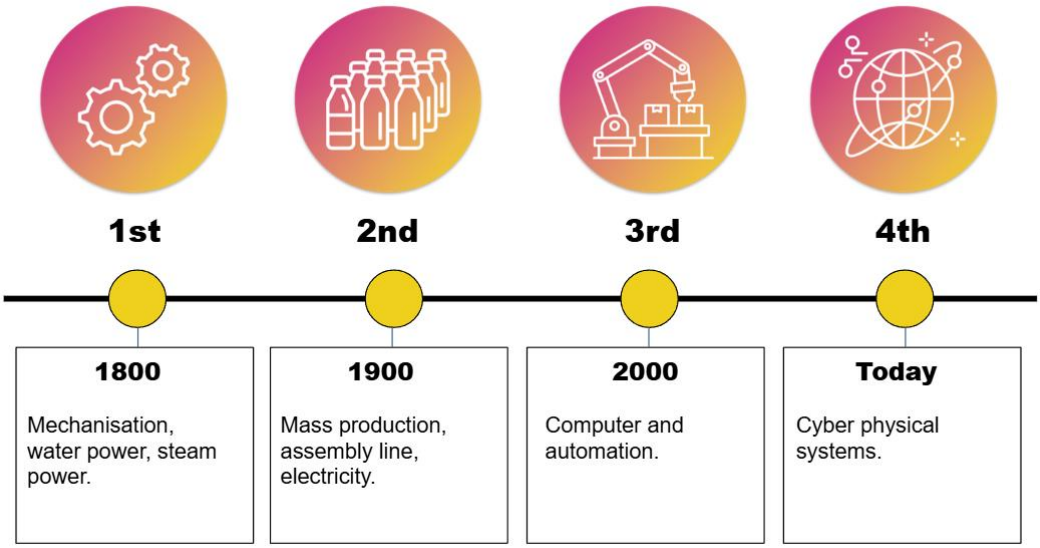
Manipulating Your Personality

Big data are exploited not only for commercial benefit but also for political ambition. The use of Facebook data in both the 2016 US Presidential Election and the Brexit election is a testament to how big data and machine learning can be exploited to change voters' minds and thereby a regime.

Lesson 5 – A timeline of industrial revolution

The technological revolution we are in will fundamentally alter the way we live, work, and relate to one another, and no-one is exactly sure how the future will look.

Hence, our response to change must be one that involves everyone in society, from the public and private space to academia and politics.



For more on this, read the Schwab text (2016) provided in the reading list for this lecturecast.

Lesson 6 – Self-Drive Cars

Displacing human drivers with self-drive cars.

Internet of Things (IoT) is a collection of devices, sensors, and services that exist in an interconnected ecosystem with the aim of sharing data within and between systems. The number of connected things has been predicted by different people, some say up 50 billion and some say 200 billion by 2020.

The development of self-driving cars is the most trendy and talked-about issue of the Industry 4.0 revolution, artificial intelligence, and machine learning. Driverless cars, like human drivers, can identify objects, interpret situations and make control-related decisions. Automotive Artificial Intelligence uses object detection and object classification algorithms of machine learning.

Lesson 7 - Living With AI

Big data, IoT and AI are transforming our lives in many ways that have short mid and long-term positive and negative impacts, many of which we are not aware of or do not exist as of yet.

Lesson 8 - Classifying Spam Email

Machine Learning is widely used to ensure safety and security of individuals in digital space. Classification of spam email is one of the everyday used cases.

Emails are coming into our inbox, and we want to create a program to identify spam emails and move these into the Junk mail folder.

Regular email (ham) should go to the inbox folder.

How can we classify our emails?

We need to create a machine learning model to help us!

We need to analyse existing emails, categorise it as spam or regular email.

To analyse refers to the process of identifying key features that makes an email as "spam" or "regular".

So, we must identify these key features & label each one (as shown in the 3 examples here).

One: Develop a model to “learn” from existing datasets (training).

Two: “Test” the model to know how good the program works (testing).

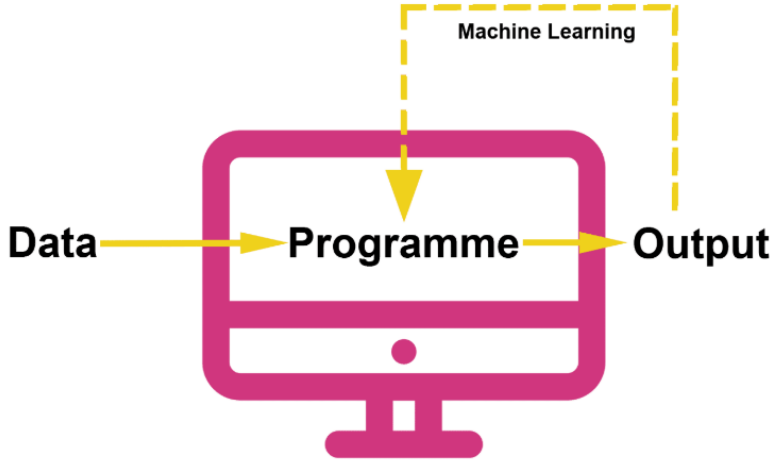
Three: “Optimise” the model.

And four: “Deploy” the model to classify each email (regular or spam).

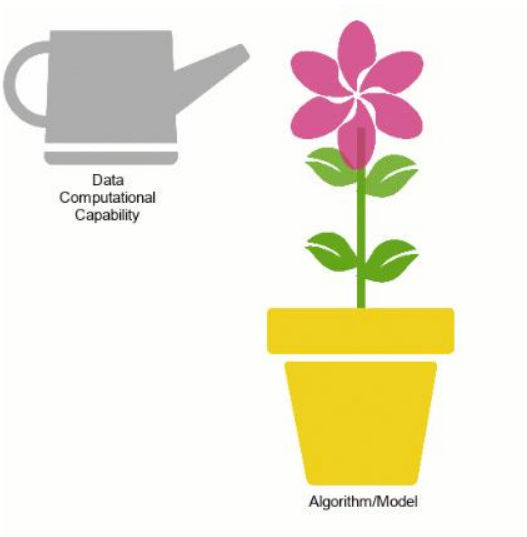
Lesson 9 - Factors Impacting Machine Learning

Machine learning algorithms, by definition, “learn” from the output.

In other words, how accurately they classify or make the right decision depends on the output.



However, the effectiveness of the algorithm, especially during the training phase will depend on the quality of the data (that is how accurately the training data was **labelled** - data that are given tags or labels, and thus made more meaningful) and the efficiency of the computational capability during the testing and deployment phase.



Accurate classification/decision that is delayed has little or no value.

Therefore, data and computational capability help to optimise an algorithm, just like water used for plant growth.

Lesson 10 - Big Data, Computational Capability and AI

Big Data

The 4 V's of big data (Volume, Velocity, Variety and Veracity) have seen exponential growth in past decades and have benefited from the adaptation and development of machine learning algorithms. However, the computational capability or velocity that is required by the machine learning algorithm still lags behind the growth of big data.

One of the key areas is image processing, live images simultaneously fed from multiple cameras (in the case of a self-drive car camera) still imposes a challenge for machine learning models in terms of classification of objects on real-time basis.

Volume: The volume of data in the last two decades has grown exponentially, (see lesson 1 – value of data). However, the cost and efficiency of storing this huge amount of data reduced and increased significantly over the last two decades.

Velocity: The velocity refers to how efficiently we transmit the data through the internet and other forms of wireless or wired networks. Again, within the last two decades, this number and type have increased many folds with the introduction of fibre optics and laser technologies.

Variety: The last two decades also witnessed a huge transformation in terms of data types or variety. Previously it was mostly structured textual data but with the advent of digital cameras and social media, there is a tsunami of audiovisual and unstructured data (free text). Interoperability within different data formats remains a big challenge for the next generation of data scientists.

Veracity: The veracity or uncertainty of big data refers to both data quality and availability. Although 2.5 quintillion bytes of data are generated every day and a big part of those data are open to public use, businesses are often unsure about the quality of the data, that is how the data was collected? What is the source of the data? etc. Such uncertainty reduces the use of the data for further analysis and thereby the value of the data. Availability of data at the right time and place is another factor. This is more prevalent in healthcare as most of the healthcare data are personal data and therefore privacy of the data needs to be ensured before any use of the data.

Computational Capability

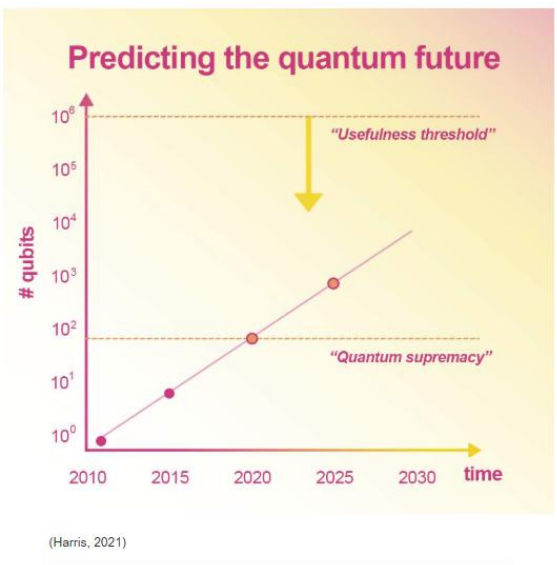
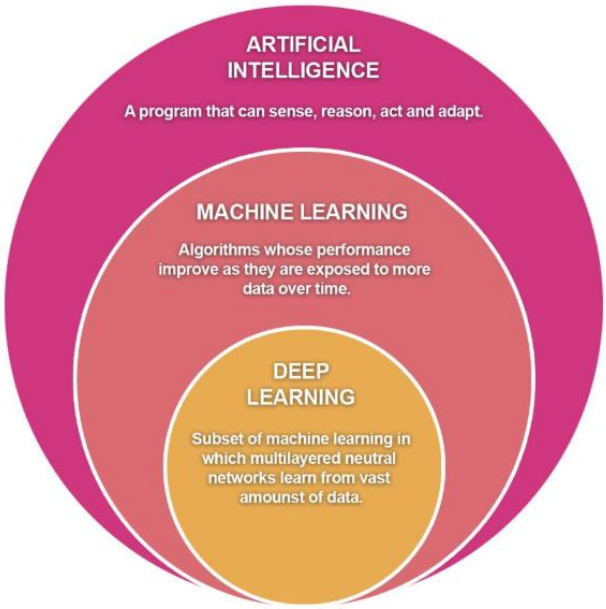


Figure 1 (Harris, 2021)

Classical computers running on CPU/GPU have limitations in terms of computing speed. The advent of quantum computing, which is capable of computing simultaneously, is deemed to be the future of computing. However, quantum computing is still in its infancy and it is predicted that the true usefulness of quantum computing will not be experienced in the next 10 years.

Artificial Intelligence

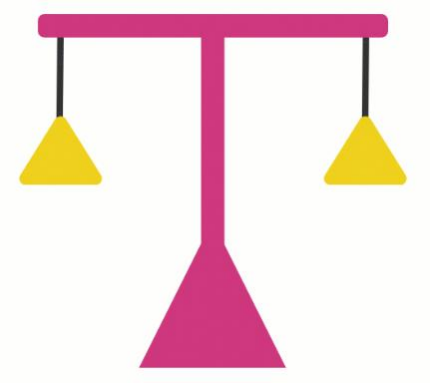


Lesson 11 - Categories of Machine Learning

Machine learning can be placed into three categories as described in this table. It should be noted that the ML algorithm in all categories requires a training dataset to train the model.

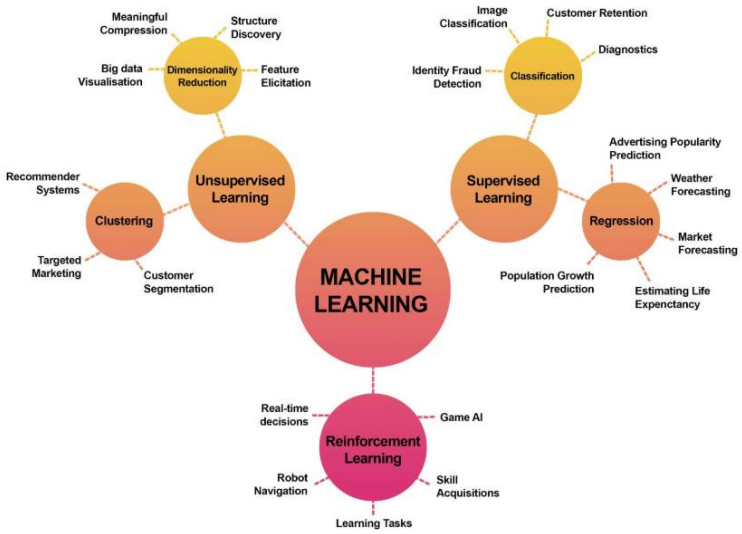
Type	Training Data	Label/Desired Output	Reward
Unsupervised	✓		
Supervised	✓	✓	
Reinforced	✓	✓	✓

The basic difference between unsupervised and other categories is that unsupervised machine learning does not have the desired output because the dataset is not labelled with class. Due to the lack of labelling of the dataset and no desired output, unsupervised algorithms do not have the capability to learn from their errors. In the other two categories, the datasets are labelled, there is the desired output, and the algorithm learns from its errors.



For reinforced learning, as the output becomes close to the desired output (in other words, fewer errors are incorporated), a reward mechanism helps to rebalance the weights and or bias, to make fewer errors in the following epoch (the number of passes of the entire training dataset the machine learning algorithm has completed).

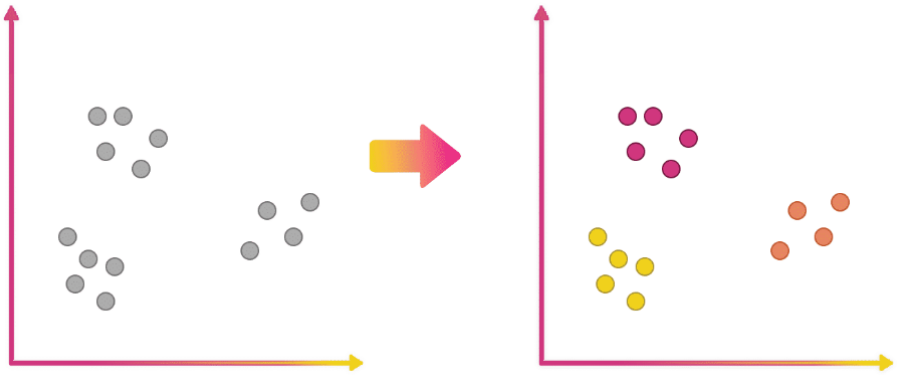
The figure below shows different categories of ML and their areas of use.



Lesson 12 - Unsupervised, Supervised and Reinforced Learning

Unsupervised Learning

In unsupervised learning, we do not have labelled data. A model can identify patterns, anomalies, and relationships in the input data using methods such as clustering.



Example of clustering

In the example below (Figure 1), we have a representation of genes from thousands of biological samples which are clustered and coloured accordingly.

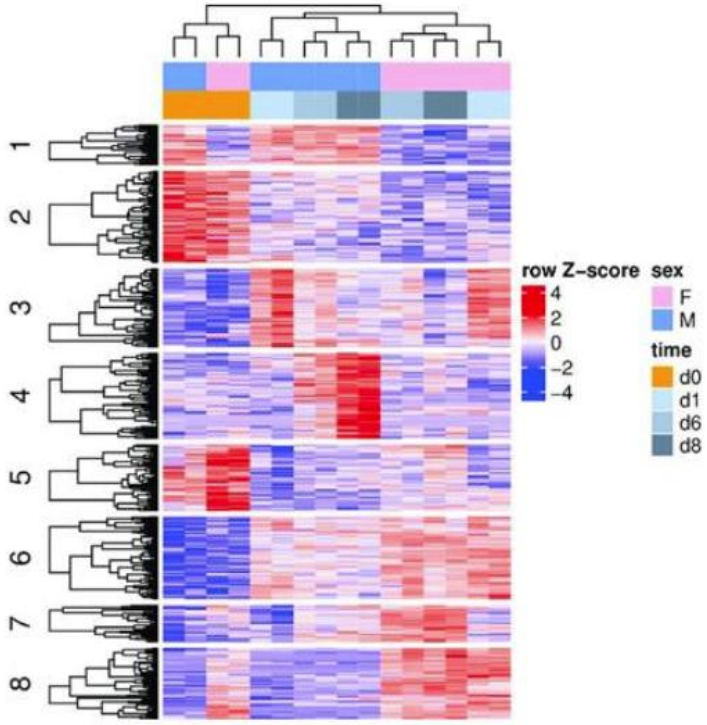


Figure 2 Clustering of Biological Samples (Pearson et al, 2019)

Supervised Learning - Regression

We can use (learn) a function f(x) to predict y when x is known. If the final value of y is real, then this is regression. In supervised machine learning, a model makes predictions or decisions based on past or labelled data (Figure 2).

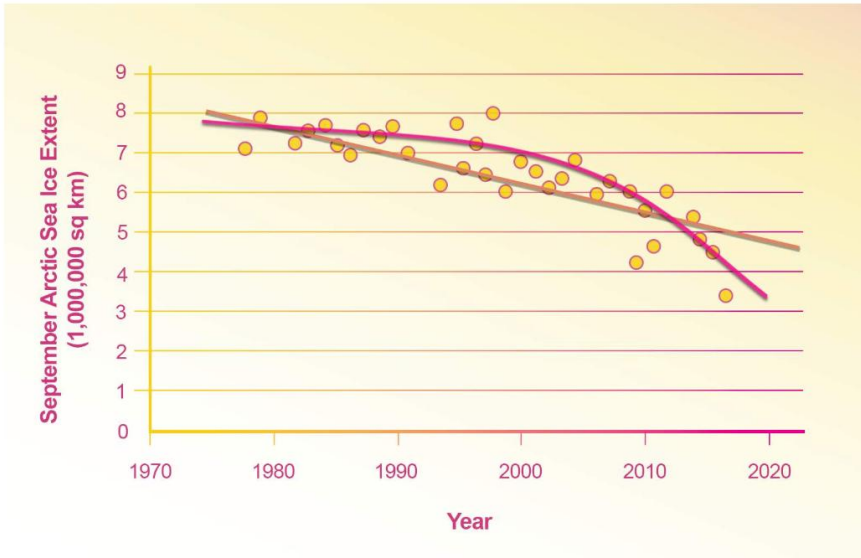


Figure 3 Prediction Model (Witt, 2013)

Supervised Learning - Classification

Given n pairs of x and y variables, we can use (learn) a function f(x) to predict y when x is known. If the final value of y is categorical in nature, then this is classification.

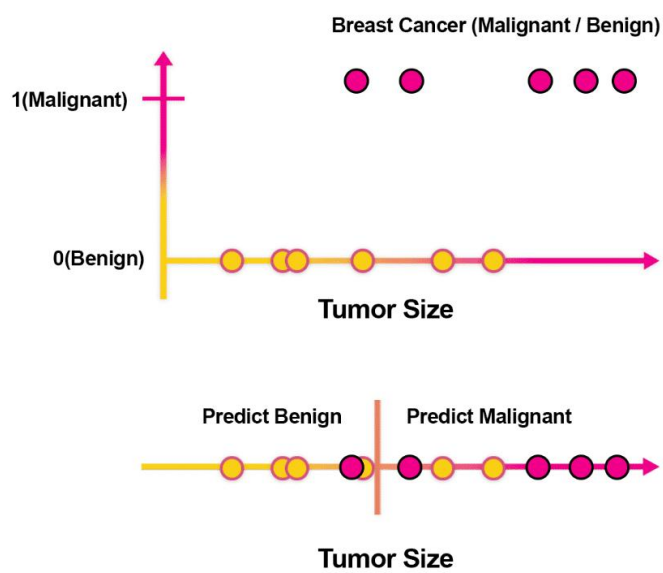
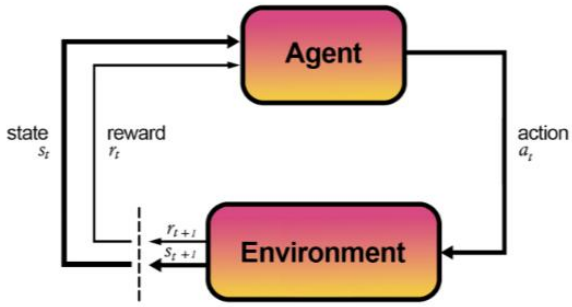
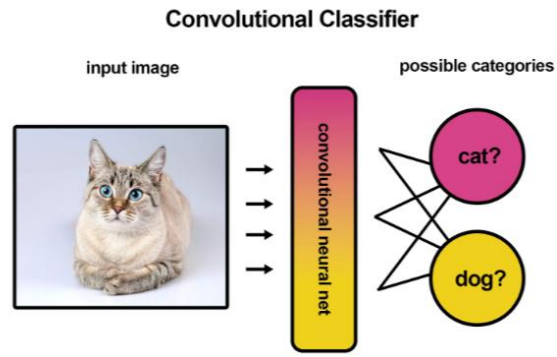


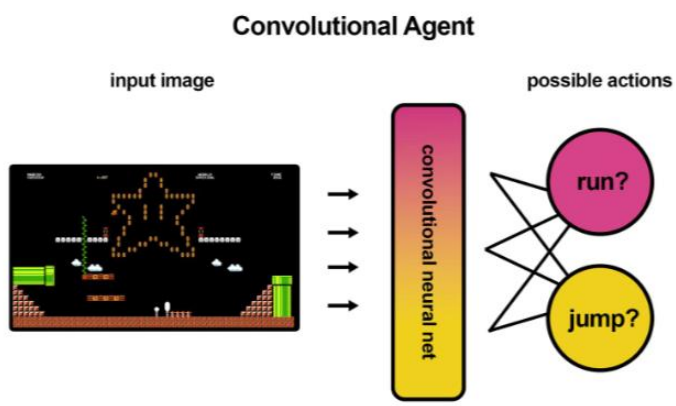
Figure 4 Based on an example by Andrew Ng

Reinforced Learning-Classification/Action

Using reinforcement learning, the model can learn based on the rewards it received for its previous action. Reinforcement learning combines artificial neural networks with a framework of reinforcement learning that helps software agents learn how to reach their goals. That is, it unites function approximation and target optimisation, mapping states and actions to the rewards they lead to.







For more information on reinforced learning, visit this wiki: https://wiki.pathmind.com/deep-reinforcement-learning

Lesson 13 - Example Applications of Machine Learning

These are some of the current applications of ML techniques and algorithms:

1. Web Search (e.g.: Google-like search engines)
2. Computational Biology (e.g.: Creating a gene network from gene expression)
3. Finance (e.g.: Customer profiling based on credit history)
4. E-commerce (e.g.: Customer targeting based on shopping habits)
5. Space exploration (e.g.: Discovery of new galaxies from telescope images)
6. Robotics (e.g.: Car manufacturing)
7. Social Network (e.g.: Cambridge Analytica and other companies that changed, or tried to change, users' behaviour and mindset)

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Proof of completion of lecture cast:

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100% COMPLETE

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- A Timeline of Industrial Revolution
- Self-Drive Cars
- Living With AI
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- Factors Impacting Machine Learning
- Big Data, Computational Capability and AI
- Categories of Machine Learning
- Unsupervised, Supervised and Reinforced Learning
- Example Applications of Machine Learning
- Check Your Knowledge

SUMMARY AND FEEDBACK

- Summary and References
- Feedback